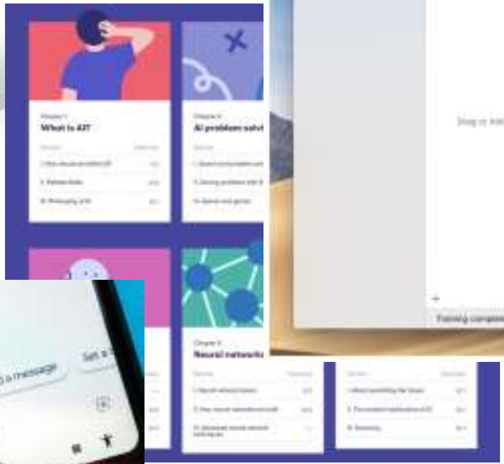
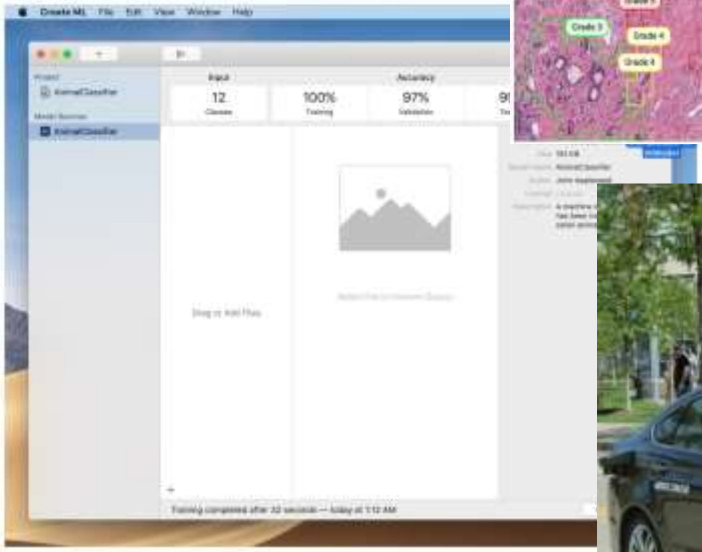
Human-AI Interaction

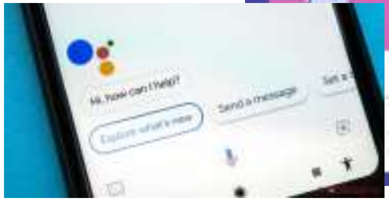
Tommaso Turchi

Adapted from:
https://elite.polito.it/2024-haii-materials/

AI is everywhere!





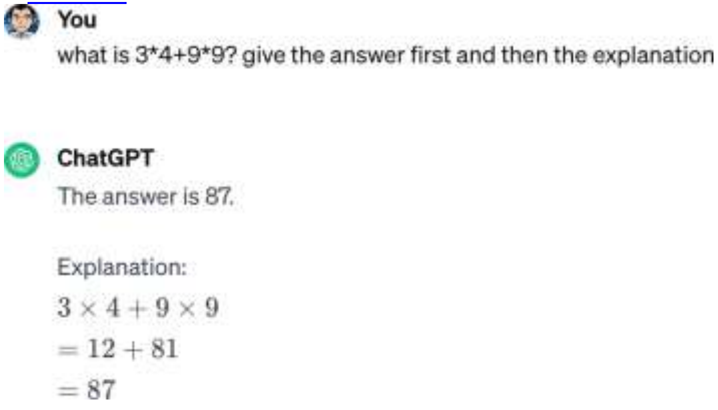


AI is everywhere!

- When it "works", it's *great*!

AI is everywhere!

- When it "fails", it does it *spectacularly*...
  - e.g., Tesla Smart Summon, https://www.youtube.com/watch?v=VbVoTK-lM00

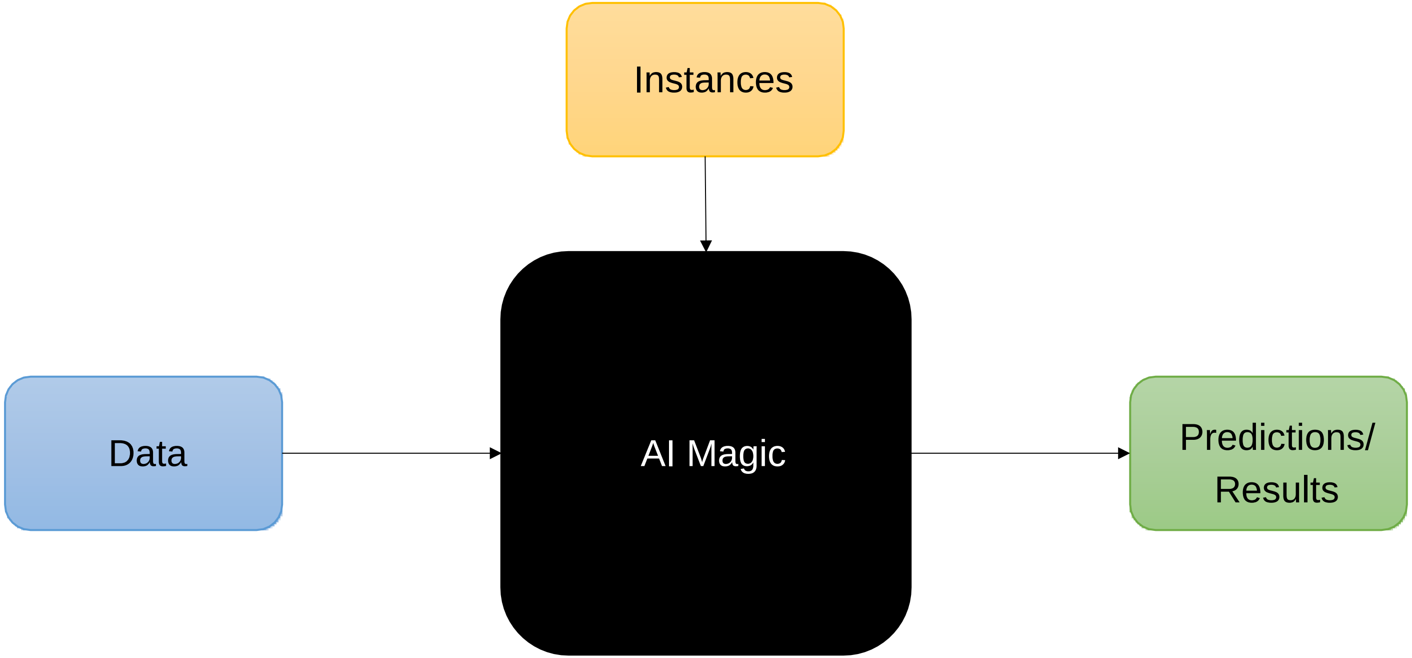


AI is everywhere!

- … and/or it is ***very*** problematic, for instance:
  - "*IBM boasted that its AI could 'outthink cancer.' Others say computer systems that read X-rays will make radiologists obsolete…*"
  - *"Systems developed in one hospital often flop when deployed in a different facility.*
    Software used in the care of millions of Americans has been shown to discriminate against minorities.
    And AI systems sometimes learn to make predictions based on factors that have less to do with disease than the brand of MRI machine used, the time a blood test is taken or whether a patient was visited by a chaplain."

    [source: https://www.scientificamerican.com/article/artificial-intelligence-is-rushing-into-patient-care-and-could-raise-risks/]
- Why?

A Possible Reason: The Typical Approach



Motivation

- Most AI/ML courses consider “user interfaces” or humans as an *afterthought*, near the end
  - several times they do not even think about "humans" ☹
  - they focus on algorithms/models, basically
- Why do not consider people from the *beginning*, and along the design, algorithmic choices, … in an *iterative* way?!

Ultimately, AI Systems Are…

- Designed by *humans*
- To solve a problem framed by *humans*
- With *humans* taking specific choices (e.g., which algorithm to use)
- Evaluated and tested by *humans*
- With an outcome for *humans* (often)
- Presented to *humans* with a user interface (sometimes)

Algorithms As The (Main) Answer?

- Algorithms are **not always** the "answer"
  - for instance: if you go to Netflix for the first time, what should it recommend you watch?
  - this is the *cold start problem*, and it is not really and fully solved
    - algorithmically speaking, at least

⇒ A **suitable** user interface is **critical** to overcome some limitations!

  - Keeping people involved and considering them since the beginning is **fundamental**!

Challenges

- How to ensure that people use AI-powered interfaces and systems with joy and trust rather than frustration and disappointment?
- How can we design and evaluate human-centered AI systems?
- How can we avoid (or minimize) problems, failures, ethical issues, … in AI systems?

People & Computers

"The two hardest problems in computer science are: (i) people, (ii), convincing computer scientists that the hardest problem in computer science is people, and, (iii) off by one errors."

Prof. Jeffrey P. Bigham, 2018

http://www.cs.cmu.edu/~jbigham/

What is Different in Interactive AI Systems?

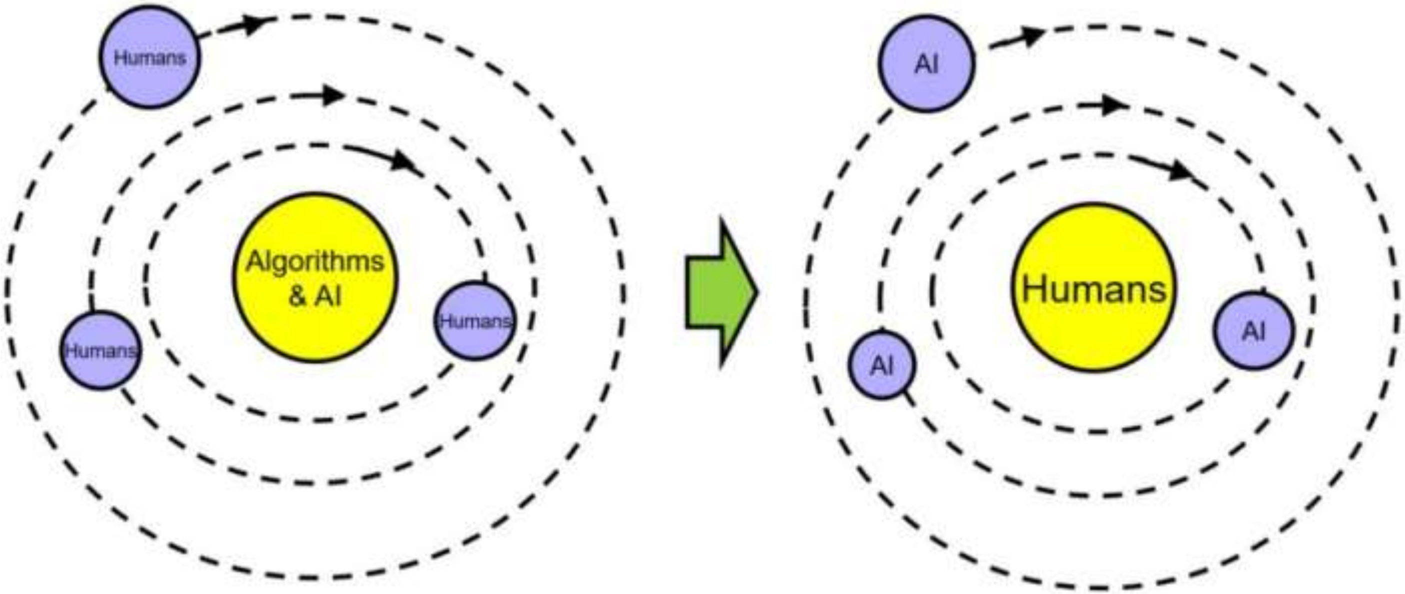
- AI-based systems are typically performed under **uncertainty**
  - often producing false positives and false negatives
  - they can be **incorrect**
- They may demonstrate unpredictable behaviors that can be *disruptive*, *confusing*, *offensive*, and even *dangerous* for users

Human-Centered AI

- Human-centered AI focuses on **amplifying**, **augmenting**, and **enhancing** human performance in ways that make AI systems **reliable**, **safe**, and **trustworthy**
- Shift from measuring **only** algorithm performance to evaluating human performance and satisfaction, with **human-centered** and participatory approaches (for evaluation, too)

Ben Shneiderman, *Bridging the Gap Between Ethics and Practice: Guidelines for Reliable, Safe, and Trustworthy Human-centered AI Systems*. ACM Transactions on Interactive Intelligent Systems, Vol. 10, No. 4, Article 26, 2020, https://doi.org/10.1145/3419764

A Paradigmatic Change



Can AI and People Really Work Together?

1. Should we design and develop <u>technology</u> that
   - automatizes people's actions, replaces humans (*Artificial Intelligence*)
   - or augments them (*Intelligent Augmentation*)?

Is there a clear winner?

2. Which kind of human-AI collaboration can we envision?
   - full human control… no human involvement?
   - full AI autonomy… no AI?
   - in which cases?

Artificial Intelligence vs. Intelligent Augmentation?

Put-That-There: Assumptions

- About speech recognition
  - why do you need to point?
  - why do you need to talk?
  - what might go wrong?
- About people
  - How would this augment them?
  - How might it fail?

Measuring Intelligence…

- How do we measure “intelligence” in AI?
- We can more accurately measure the **joint performance** of a task for a human together with AI
  - How human and machine together can achieve something that a human or the machine alone cannot (or would do worse)
- **Collaboration is the key**

Humans or AI: Who Should Have The Conn?

- It is essential that people *feel* in control of their lives and surroundings
- When we “put intelligence” in things, people should:
  - be **comfortable** with the actions made by the intelligent system
  - **understand** why some actions are happening
  - **trust** the intelligent system
- *Automation* is typically met with *resistance*
  - however, it can reduce the workload and allow to complete dangerous tasks
- We should avoid (and consider) over-exaggerated expectations
  - *claimed*: “we have reach human-parity in speech recognition!”
  - *pre-assumed*: "I can speak with it, it understands my words, THUS it has full language understanding”

Full Self-Driving Cars

- Who should be in control? When? What can go wrong? Why?

source:
https://www.youtube.com/watch?v=tlThdr3O5Qo

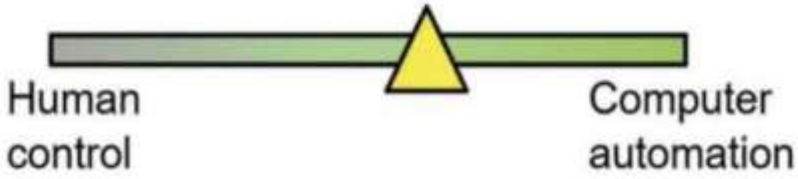
Why Not Automating WHILE Augmenting?

- Remember: collaboration, human-centered AI, …
- “Human-centered AI focuses on **amplifying**, **augmenting**, and **enhancing** human performance in ways that make AI systems **reliable**, **safe**, and **trustworthy**”

- Examples from the real world?

Human-Centered AI Framework

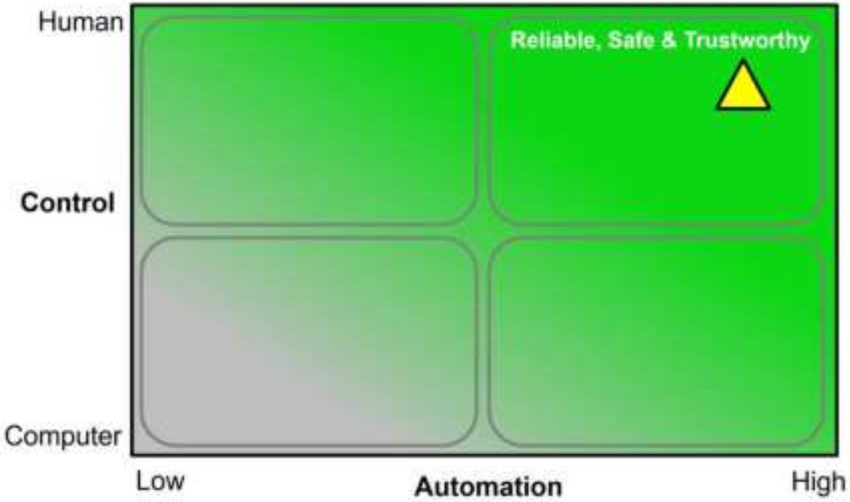
- Until now, we have seen human control in *contraposition* to automation
- This is also very typical
  - e.g., think about the levels of driving automation



Ben Shneiderman, *Human-Centered Artificial Intelligence: Reliable, Safe & Trustworthy*. 2020. International Journal of Human-Computer Interaction. https://doi.org/10.1080/10447318.2020.1741118

Human-Centered AI Framework

- What if do we move to a 2D framework?



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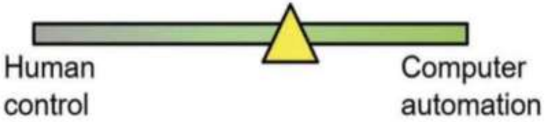
AI Goal: Emulation



- **Automation:** carrying out requirements anticipated at design time

vs.

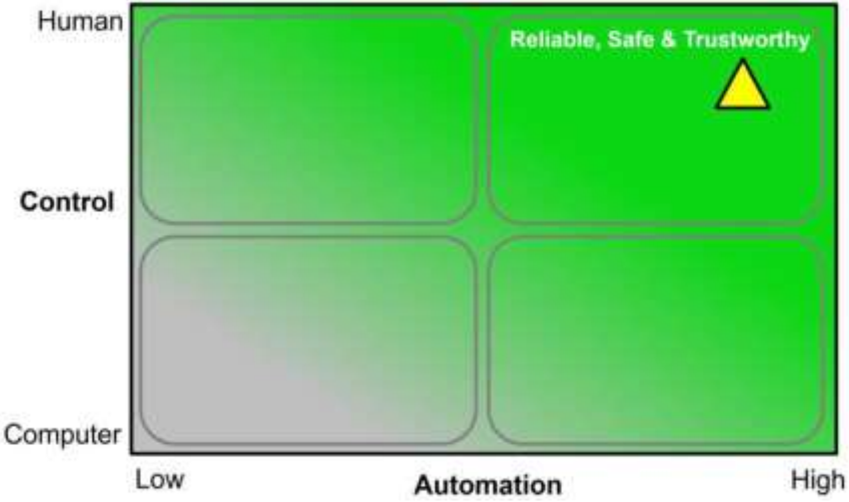
- **Autonomy:** supporting unanticipated goals and emergent behaviors based on new sensed data

- Emulation belief: AI systems can be **autonomous**, **independent**, **capable of setting goals**, **self-directed**, and **self-monitoring**
  - teammates, partners, collaborators
  - embodied intelligence through humanoid robots

Ben Shneiderman, *Design Lessons From AI's Two Grand Goals: Human Emulation and Useful Applications*. 2020. IEEE Transactions on Technology and Society. https://doi.org/10.1109/TTS.2020.2992669

AI Goal: Application

- Development of widely used products and services by applying AI methods, fulfilling some design goals
- High-levels of human control and high levels of computer automation
- HCI methods applied to identify needs and design goals



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Interactive Human-Centered AI

- “An Artificial Intelligence that enables **interactive exploration** and **manipulation** in real time and is designed with a clear purpose for **human benefit** while being **transparent** about who has **control over data and algorithms**.”

  [A. Schmidt. *Interactive Human Centered Artificial Intelligence: A Definition and Research Challenges.* AVI 2020]

- **Interaction** as a key for the collaboration between human and AI: the channel for the AI algorithm to support the users and for the users to control the algorithm
  - **Observable Intelligence**

Properties of Interactive Human-Centered AI

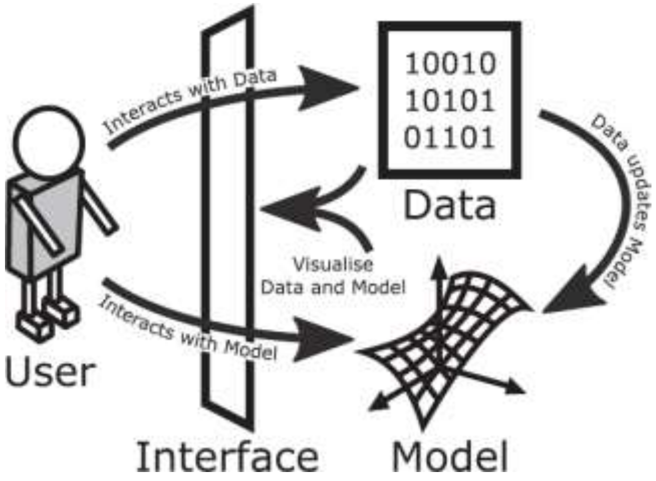
1. Individuals can **interact in real time** with the **algorithms**, **models**, and **data** and can manipulate and control all relevant parameters
2. The impact of changes and **manipulations** made by the user can be **observed in real time**
3. Individuals can **interactively explore** why and how **specific decisions** are made and find out how changes in the parameters, data and models impact outcomes
4. In fast processes **the speed can be reduced to allow interaction,** interventions and manipulations

Properties of Interactive Human-Centered AI

5. It states clearly **how humans can benefit** form the AI
6. It explains **what risks the AI poses** for individuals as well as on societal level
7. It is visible **who has control** on the AI, who has the **power over data, models, algorithms**
8. It is visible **what data, knowledge base, and information** is used **to create and inform** the AI

Understanding Systems

- Not only opening code
  - not understandable for many people
  - little insights even for AI specialists
- Design **goals** and **functionality** that the AI system is supposed to support



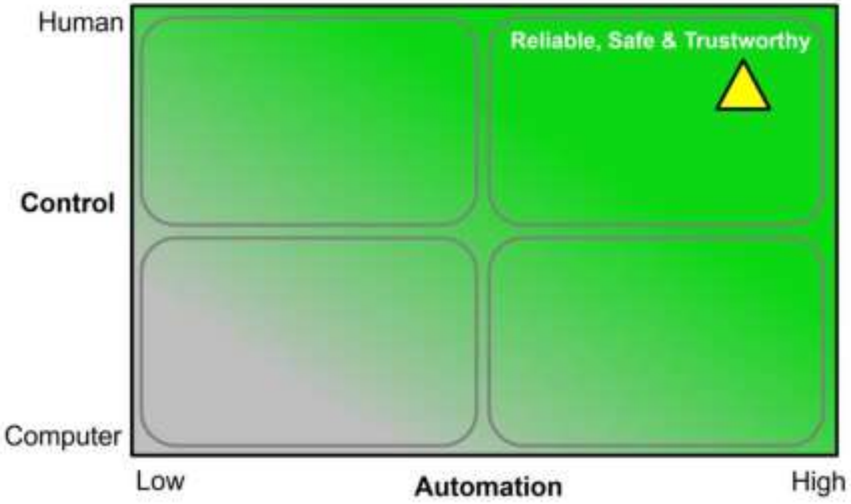
John J. Dudley and Per Ola Kristensson. 2018. A Review of User Interface Design for Interactive Machine Learning. *ACM Trans. Interact. Intell. Syst.* 8, 2, Article 8 (June 2018), https://doi.org/10.1145/3185517

Distinct Interface Elements

- Output review
  - visualizing (a selection of) output samples
- Feedback assignment
  - corrections and/or creation of new samples
- Model inspection
  - Evaluation of the model quality
- Task overview
  - Contextualizing tasks to help the assessment by naive users
- Anything else?

Human-Centered AI Framework

- What if do we move to a 2D framework?



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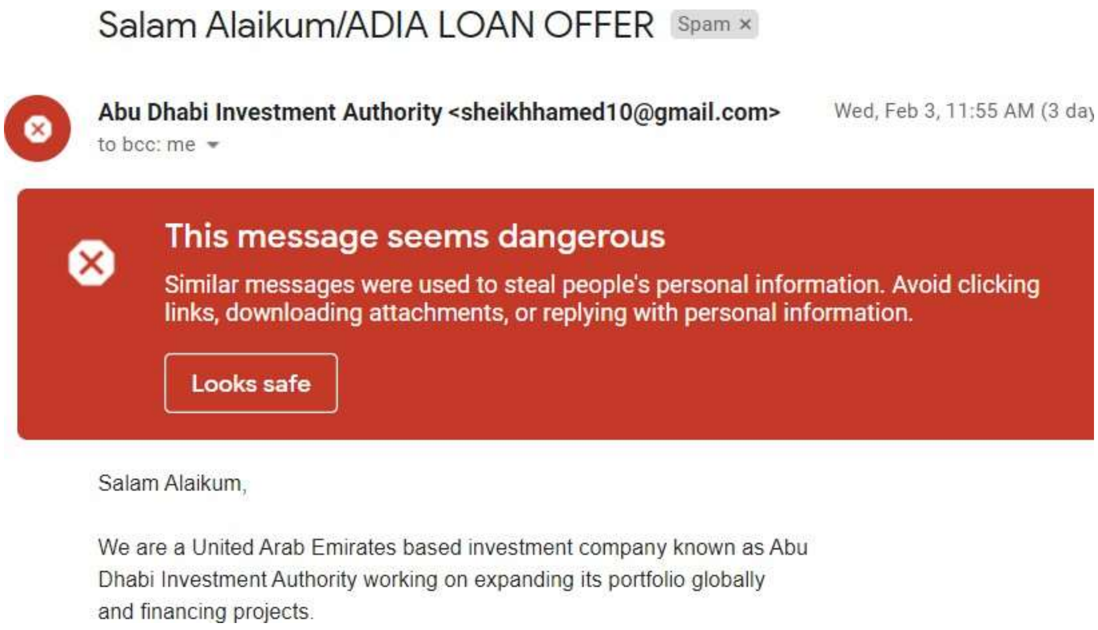
Gmail spam filter

- No input needed
- User can override decisions already taken by the system



Google Nest thermostat

- Initial set up
- Automatic learning (very sensitive in the first two weeks, much less after)
- Continuous adjustments in time

https://www.youtube.com/watch?v=20367DapHlc

Google Nest thermostat

- Automatic learning (very sensitive in the first two weeks, much less after)
- Continuous adjustments in time

Pattern of temperature changes	How it changes your thermostat's schedule
Two weekdays in a row (Monday and Tuesday)	All weekdays (Monday to Friday)
Same day two weeks in a row (two Mondays in a row)	That day of the week (every Monday)
Two weekend days in row (Saturday and Sunday)	All weekend days (Saturday and Sunday)
Two days in a row including a weekday and a weekend (Friday and Saturday)	All seven days of the week (Monday to Sunday)

Amazon Alexa

- Vocal commands in natural language
- Vocal responses and actions

https://www.youtube.com/watch?v=Ymewnb3gJJQ

Amazon Alexa

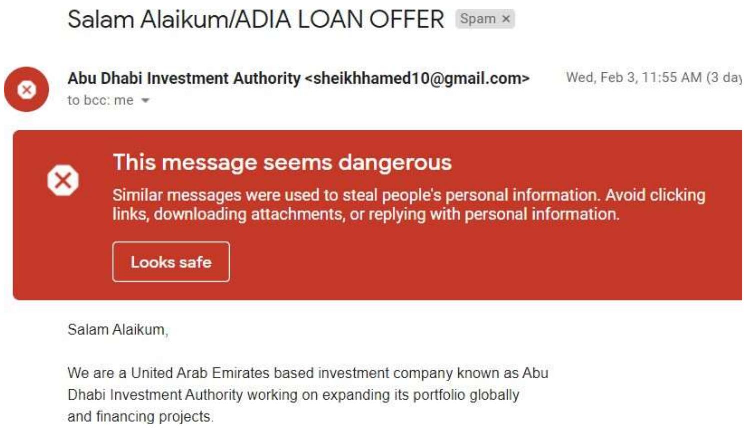
- *Sorry, I'm having problems in understanding you right now…*

https://www.youtube.com/watch?v=XQCHoKAq9xA

Google Home

https://www.youtube.com/watch?v=e2R0NSKtVA0

AI-based systems as smart tools



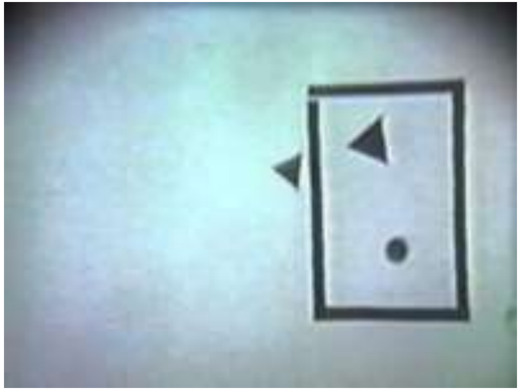
AI-based systems as smart tools

- Digital technologies are Cognitive Artifacts: physical objects designed to display or operate about information for enhancing human cognition (Norman, 1991; Hutchins, 2002)
- *Cognitive Artifacts + Artificial Intelligence = smart tools*
  - look like standard GUIs
  - aim to alleviate some tasks by acting autonomously
  - users are meant to be in control through the interface
  - might be confusing in terms of autonomy vs control because of probabilistic model

AI-based systems as artificial companions

AI-based systems as artificial companions

- Interaction with intelligent systems based on the metaphor of human-human interaction
- Human beings are coded to adopt an intentional stance
  - a tendency to anthropomorphize tools (e.g. Heider-Simmel illusion): yet, that does not imply that we actually believe that tools are intelligent (Reeves and Nass, 1996)
- There is evidence that anthropomorphic features increases UX
  - anthropomorphic features increase trust in an automated car (Waytz, Heafner, and Epley 2014)
  - expression of emotions improves efficacy in collaborative decision making tasks (de Melo, Gratch, and Carnevale 2015)

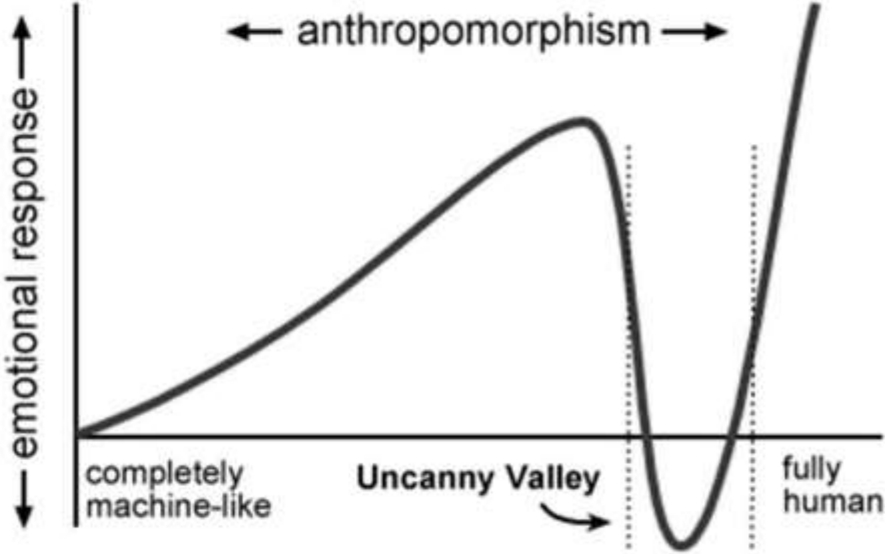


Heider-Simmel Illusion (1944)

AI-based systems as artificial companions

- Yet, in the long term, UX can worsen
  - the presence of an anthropomorphized helper reduces enjoyment in games (Kim et al., 2016)
  - over-reliance and over-trust can in the long term bring to security and safety issues (Chung et al., 2017)
- Small aspects can induce larger and unwanted effects, e.g. people attribute negative stereotypes to female-presenting chatterbots more often than they do to male-presenting chatterbots (Brahnam & De Angeli, 2012)
- Keep attention to the **Uncanny Valley**!

AI-based systems as artificial companions



Summing up

Smart Tools

- Smarter but less predictable than objects
- Opaque mental model
- Principles of Interaction Design
- New principles to manage AI

Artificial Companions

- Almost but not like humans
- Encourage social attribution
- Uncanny valley
- Different principles

AI: Risks, Benefits, and User Tolerance

What is Different in Interactive AI Systems?

- AI-based systems are typically performed under **uncertainty**
  - often producing false positives and false negatives
- They may demonstrate unpredictable behaviors that can be *disruptive*, *confusing*, *offensive*, and even *dangerous* for users

Low-stake Examples

- **Relevance** errors
  - Airbnb suggesting "fun local activities" when you are traveling for a funeral
  - Exercise app suggesting "time to get up and walk!" when you are seated on a long car trip
- **Multiple** users, **similar** input
  - Use Spotify to play 1970s pop jams at a thematic party
  - Use Spotify to play your favorite study jams at home
  - Use Spotify to hate-listen to <insert here an artist you dislike> with your roommate

What music should Spotify recommend this account play?

What Are The Stakes For AI Failure?

User: low stakes

- AI feature is annoying or interrupting
- AI feature is often wrong
- AI feature is useless

User: high stakes

- AI causes active harm (e.g., recidivism prediction or hiring prediction)
- AI reveals information someone wanted kept private
- AI shows offensive content

Product/Service organization

- Users stop using your app/service because of poor AI performance
- Bad press or legal troubles
- Bad reviews discouraging others from using the app/service

Traditional Guidelines and AI

- AI-based systems can also violate established usability guidelines of traditional user interface design
  - for instance: consistency or error prevention
- Many AI components are inherently **inconsistent**
  - they may respond differently to the same text input over time (e.g., autocompletion systems suggesting different words after language model updates)
  - or behave differently from one user to the next (e.g., search engines returning different results due to personalization)

What is an AI-based System?

- Artificial intelligence (AI) refers to systems that display intelligent behaviour **by analysing their environment** and **taking actions** – with some degree of **autonomy** – to achieve specific goals.

AI for Europe, COM/2018/237 https://www.europeansources.info/record/communication-artificial-intelligence-for-europe/

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Recognition

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Recognition

Prediction

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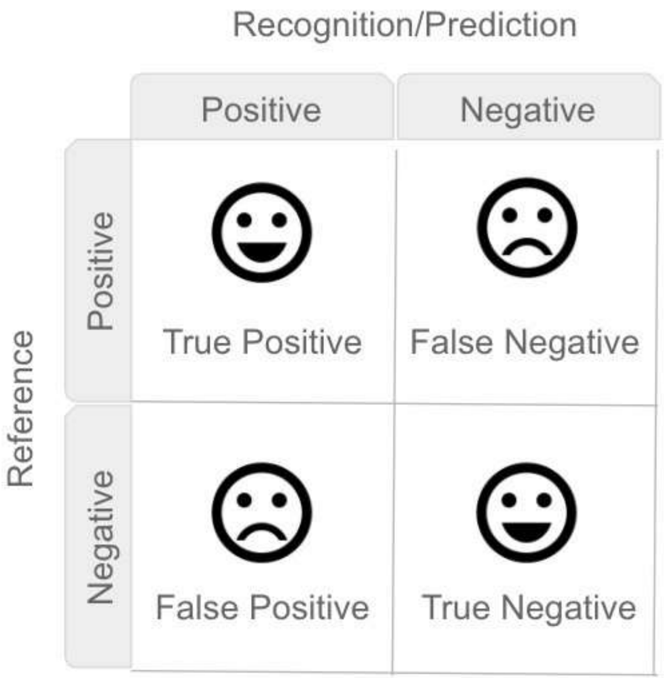
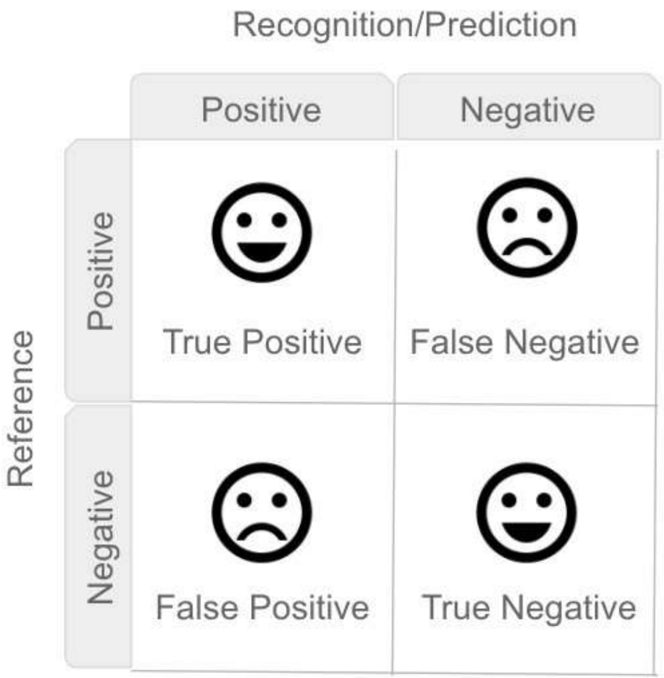
Optimizing for Precision vs. Optimizing for Recall



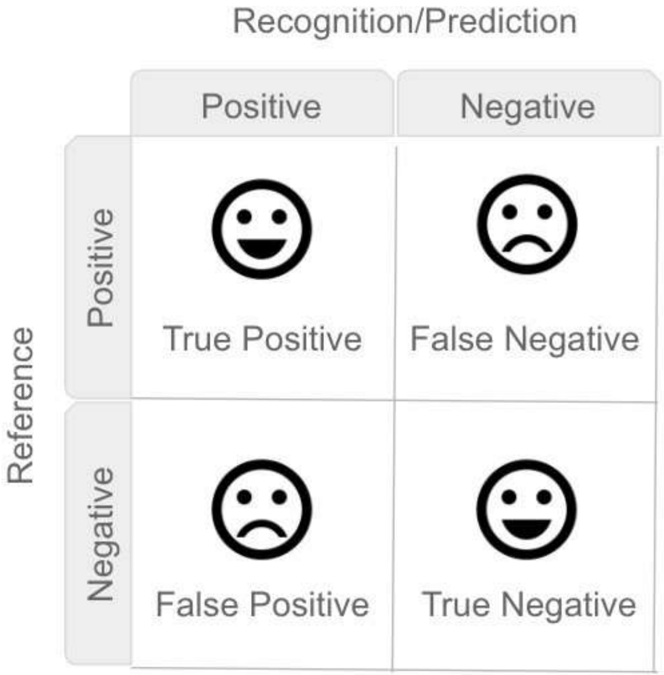
Optimizing for Precision vs. Optimizing for Recall



PRECISION =

RECALL =

Optimizing for Precision vs. Optimizing for Recall

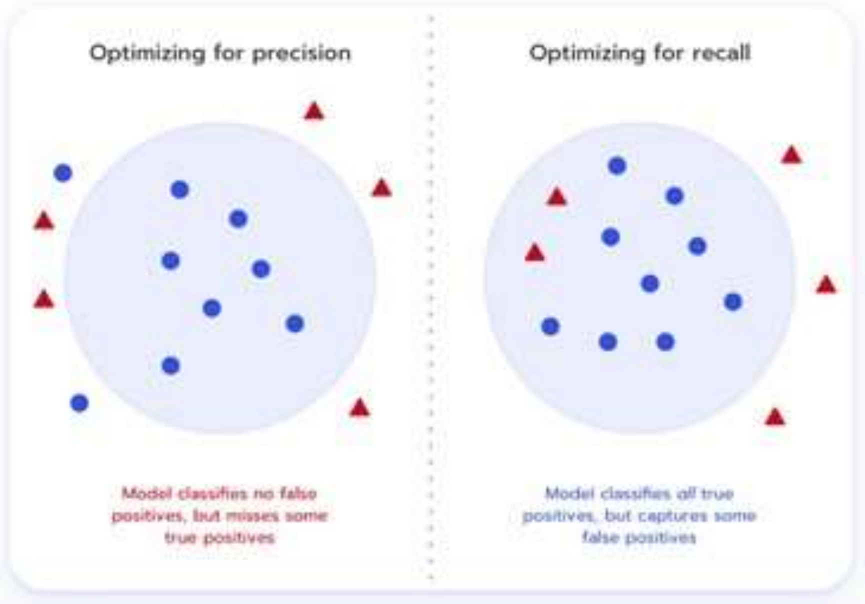


PRECISION = TP / (TP + FP)

RECALL = TP / (TP + FN)

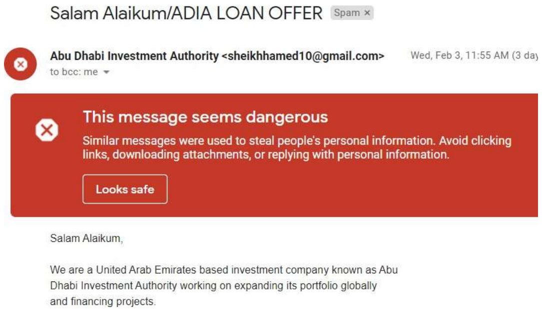
Optimizing for Precision vs. Optimizing for Recall

The worst thing is a false alarm



The worst thing is missing a positive

Should we optimizing for precision or recall?



1. Gmail spam filter

2. Google nest

3. Alexa

4. Jibo

How Can We Design Interactive AI Systems?

- By shifting from measuring **only** algorithm performance to evaluating human performance and satisfaction, with **human-centered** and participatory approaches (for evaluation, too)
- Deciding when "to AI" and when "not to AI"
- Understanding when to automate (i.e., replace the user) and when to augment users' capabilities
- Balancing the uncertainty of AI systems with proper expectations and feedback

"To AI or not to AI?"

- After identifying **user needs** and understanding *how* you can solve each of those needs
- Ask yourselves: can AI solve the user need in a unique way? Why?

AI probably better	AI probably **not** better
❑ The core experience requires recommending different content to different users.	❑ The most valuable part of the core experience is its predictability regardless of context or additional user input.
❑ The core experience requires prediction of future events.	❑ The cost of errors is very high and outweighs the benefits of a small increase in success rate.
❑ Personalization will improve the user experience.	❑ Users, customers, or developers need to understand exactly everything that happens in the code.
❑ User experience requires natural language interactions.	❑ Speed of development and getting to market first is more important than anything else, including the value using AI would provide.
❑ Need to recognize a general class of things that is too large to articulate every case.	❑ People explicitly tell you they don't want a task automated or augmented.
❑ Need to detect low occurrence events that are constantly evolving.	
❑ An agent or bot experience for a particular domain.	
❑ The user experience doesn't rely on predictability.	

source: https://pair.withgoogle.com/worksheet/user-needs.pdf source: https://developer.apple.com/design/human-interface-guidelines/machine-learning/overview/roles/

"Human-centered AI focuses on amplifying, augmenting, and enhancing human performance in ways that make systems **reliable**, **safe**, and **trustworthy**"

AI Features Meet Users

- User *tolerance* to AI features depends on the role(s) of the feature
- **Critical** or **Complementary**
  - if a system can still work without the feature that AI enables, AI is complementary
- **Proactive** or **Reactive**
  - Proactive: it provides results without people requesting it to do so
  - Reactive: it provides results when people ask for them or when they take certain actions
- **Visible** or **Invisible**
- **Dynamic** or **Static**
  - how features evolve over time

User Tolerance: Critical or Complimentary

- In general, the more **critical** an app feature is, the more people *need* accurate and reliable results
- On the other hand, if a **complementary** feature delivers results that are not always of the highest quality, people *may* be more forgiving
- Examples
  - Face ID -> critical or complementary?
  - Word suggestions (on smartphones keyboards) -> critical or complementary?
  - What happens if they fail?

User Tolerance: Proactive or Reactive

- **Proactive** features can prompt new tasks and interactions by providing unexpected, sometimes serendipitous results
- **Reactive** features typically help people as they perform their current task
- Because people *do not ask* for the results that a proactive feature provides, they may have *less* tolerance for low-quality information
  - such features have more potential to be *annoying*

User Tolerance: Proactive or Reactive

- Proactive features can be helpful
  - in small amounts
  - at the "right" moment
  - if they are easy to dismiss

User Tolerance: Visible or Invisible

- People's impression of the **reliability** of results can differ depending on whether a feature is *visible* or *invisible*
- With a **visible** feature, people form an opinion about the feature's reliability as they choose from among its results
- It is *harder* for an **invisible** feature to communicate its reliability — and potentially receive *feedback* — because people may not be aware of the feature at all
- Examples?

User Tolerance: Dynamic or Static

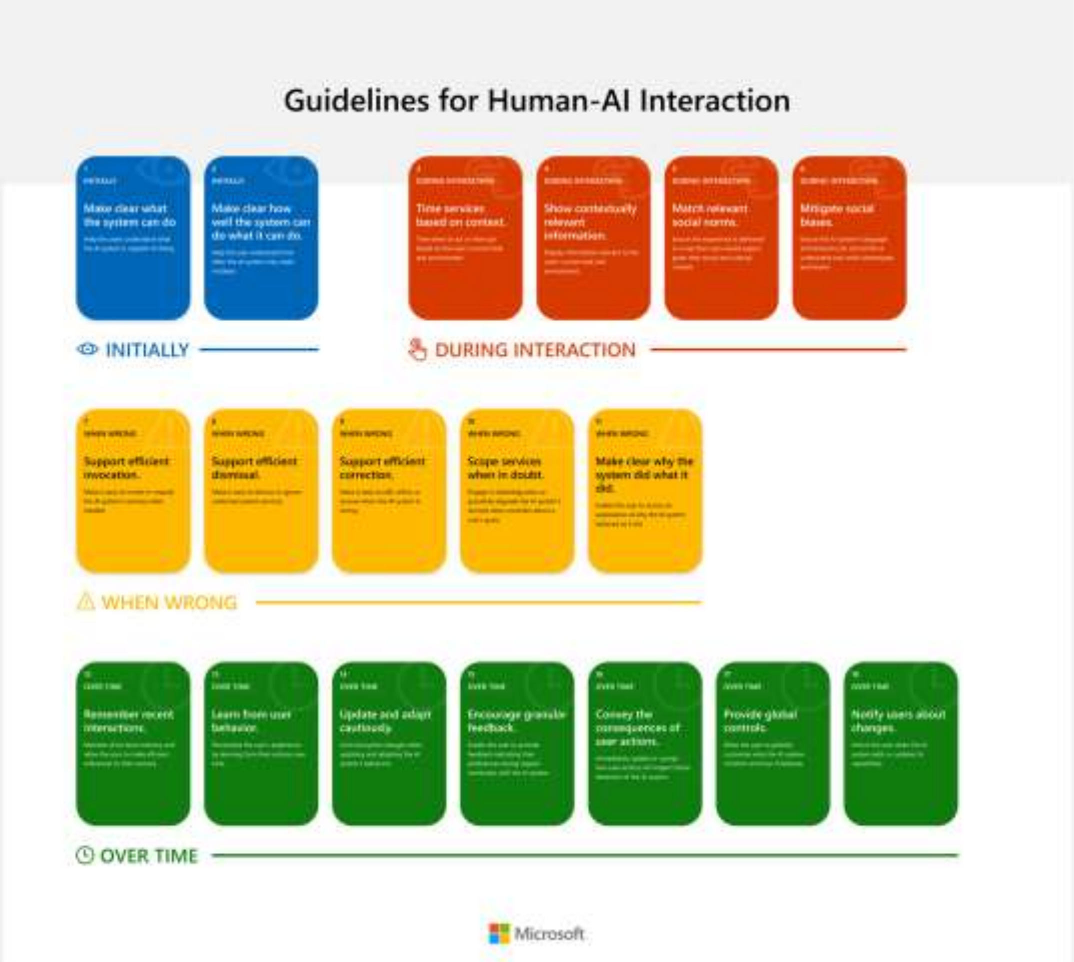
- **Dynamic** features are those that improve as people interact with the system
  - e.g., face recognition for unlocking the phone
- **Static** features *optionally* improve with a new system update
  - e.g., the quality of face recognitions in the photo library on a smartphone
- Such improvements affect other parts of the user experience
  - dynamic features often incorporate some forms of *calibration* and *feedback* (either implicit or explicit)
  - static features may not
- Depending on the feature, such updates can modify the perceived reliability, safety, and/or trustworthiness of a system

User Tolerance To Give Feedback

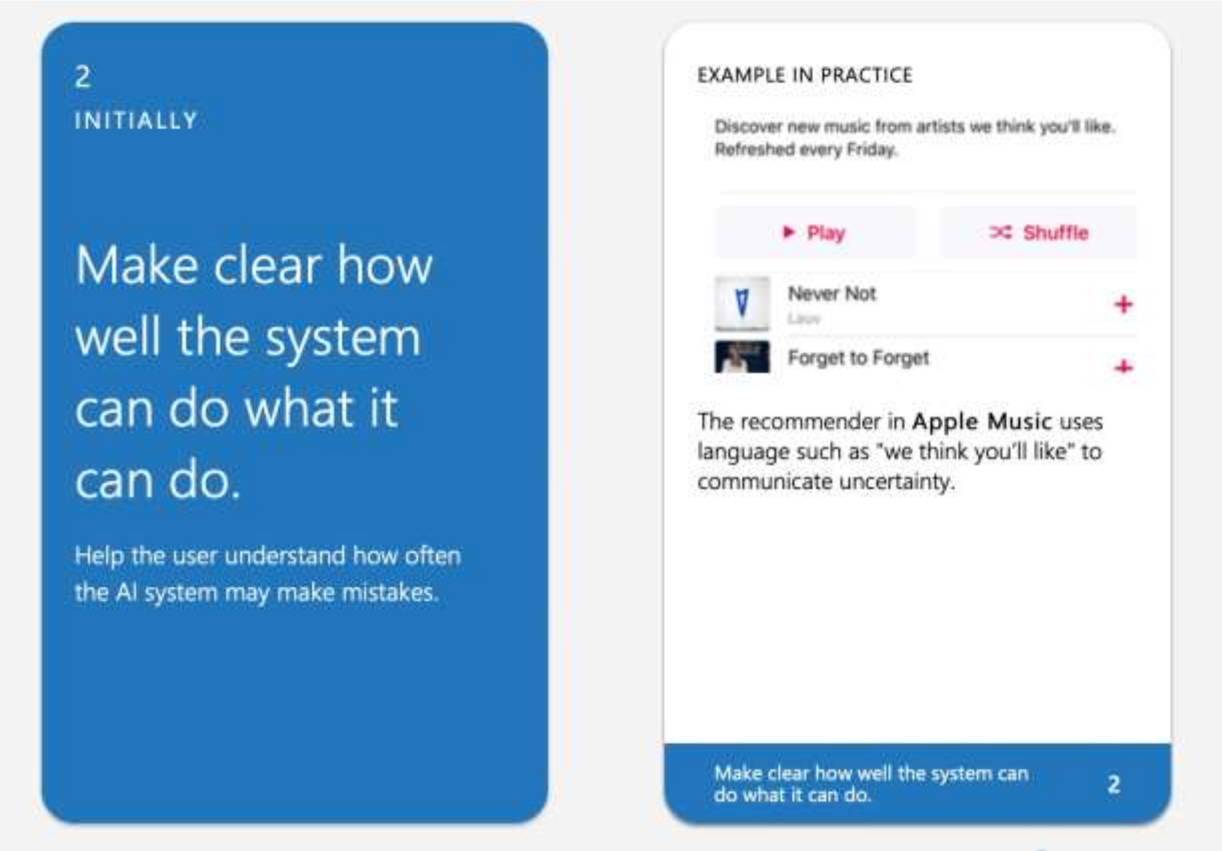
- Do not *overuse* feedback requests or users will get annoyed
  - People would not like to feel like the AI is so stupid that it needs their help
- Save for **high stakes** failure, is possible

Choosing the People+AI Path

Guidelines for mitigating risks, increasing tolerance, and highlighting benefits



By Microsoft Research: https://www.microsoft.com/en-us/research/project/guidelines-for-human-ai-interaction/


2
INITIALLY
Make clear how well the system can do what it can do.
Help the user understand how often the AI system may make mistakes.
EXAMPLE IN PRACTICE
Discover new music from artists we think you'll like. Refreshed every Friday.
Play
Shuffle
Never Not
Forget to Forget
The recommender in Apple Music uses language such as "we think you'll like" to communicate uncertainty.
Make clear how well the system can do what it can do.
2

6

DURING INTERACTION

Mitigate social biases.

Ensure the AI system's language and behaviors do not reinforce undesirable and unfair stereotypes and biases.

EXAMPLE IN PRACTICE

The predictive keyboard for **Android** suggests both genders when typing a pronoun starting with the letter "h."

Mitigate social biases. 6

9

WHEN WRONG

Support efficient correction.

Make it easy to edit, refine, or recover when the AI system is wrong.

EXAMPLE IN PRACTICE

When **Bing** automatically corrects spelling errors in search queries, it provides the option to revert to the query as originally typed with one click.

Support efficient correction. 9

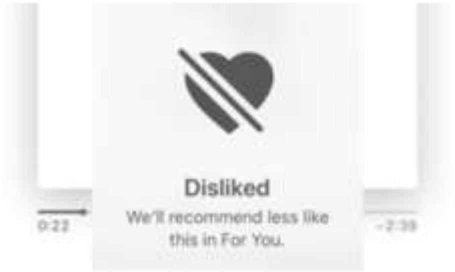
16

OVER TIME

Convey the consequences of user actions.

Immediately update or convey how user actions will impact future behaviors of the AI system.

EXAMPLE IN PRACTICE



Upon tapping the like/dislike button for each recommendation in **Apple Music**, a pop-up informs the user that they'll receive more/fewer similar recommendations.

Convey the consequences of user actions. 16

Other Guidelines

- Google's People+AI Guidebook: https://pair.withgoogle.com/guidebook/
- Apple's Human Interface Guidelines for Machine Learning: https://developer.apple.com/design/human-interface-guidelines/machine- learning/
- Microsoft's Human-AI eXperience Toolkit: https://www.microsoft.com/en- us/haxtoolkit/